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Institution: Federal University of Petroleum Resources, Effurun
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Levels: 200

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Paper 1 of 4 — EMT 221/EVS 223 Introductory Biochemistry 2022/2023

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF ENVIRONMENTAL MANAGEMENT AND TOXICOLOGY

SECOND SEMESTER 2022/2023 SESSION DEGREE EXAMINATION

EVS 223/EMT 221: INTRODUCTORY BIOCHEMISTRY II (2UNITS)

TIME ALLOWED: 2HRS

INSTRUCTION: ANSWER QUESTION ONE AND ANY OTHER THREE QUESTIONS.

QUESTION 1

- Describe the two main categories of metabolism. How do they differ from each other?
- Compare and contrast gluconeogenesis with glycolysis, highlighting key differences.
- What is the pentose phosphate pathway (PPP), and where does it occur in the cell?

QUESTION 2

- Explain the initial steps involved in the efficient extraction of energy from fatty acids, which is achieved through a repetitive four-step process.
- True or False: Is the ATP yield higher when one mole of glucose undergoes aerobic oxidation compared to the oxidation of one mole of palmitoyl-CoA?

QUESTION 3

- Argue for or against the statement, "Fatty acid biosynthesis is essentially a reversal of the beta-oxidation pathway."

QUESTION 4,

- In cases of prolonged starvation lasting several weeks, what does the body resort to as a last-ditch source of fuel or energy?
- Provide a description of a typical enzyme-catalyzed transamination reaction.
- In which organ of the body does the reaction mentioned in (b) primarily occur?
- What is a significant requirement for excreting excess ammonia in the body?
- What is the primary pathway responsible for addressing the need mentioned in (d) above?
- Excess ammonia is excreted through three main processes:
 - Direct excretion, as in ammonia,
 - Conversion to urea as observed in humans,
 - Conversion to uric acid as seen in birds.

QUESTION 5

- When exposed to xenobiotics, what are the primary toxic effects that may occur in the body?
- Delve into the Phase 2 reactions involved in the metabolism of xenobiotics and discuss their significance in detoxification.

Paper 2 of 4 — EMT 221

Level 200 • Second Semester • PastQuestion

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Paper 3 of 4 — EMT 221 Notes

Level 200 • Second Semester • Handout

1

METABOLISM

Metabolism is a highly coordinated cellular activity in which many multienzyme systems (metabolic pathways) cooperate to (1) obtain chemical energy by capturing solar energy or degrading energy-rich

nutrients from the environment; (2) convert nutrient molecules into the cell's own characteristic molecules, including precursors of macromolecules; (3) polymerize monomeric precursors into macromolecules: proteins, nucleic acids, and polysaccharides; and (4) synthesize and degrade biomolecules required for specialized cellular functions, such as membrane lipids, intracellular messengers, and pigments.

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removal, transfer, or addition of a particular atom or functional group. The precursor is converted into a

product through a series of metabolic intermediates called metabolites. The term intermediary metabolism is often applied to the combined activities of all the metabolic pathways that interconvert

precursors, metabolites, and products of low molecular weight.

Catabolism is the degradative phase of metabolism in which organic nutrient molecules (carbohydrates,

fats, and proteins) are converted into smaller, simpler end products (such as lactic acid, CO₂, NH₃).

Catabolic pathways release energy, some of which is conserved in the formation of ATP and reduced

electron carriers (NADH, NADPH, and FADH₂); the rest is lost as heat. In anabolism, also called biosynthesis, small, simple precursors are built up into larger and more complex molecules, including

lipids, polysaccharides, proteins, and nucleic acids. Anabolic reactions require an input of energy, generally in the form of the phosphoryl group transfer potential of ATP and the reducing power of NADH, NADPH, and FADH₂ (Fig. 1).

Some metabolic pathways are linear, and some are branched, yielding multiple useful end products from a single precursor or converting several starting materials into a single product. In general, catabolic pathways are convergent and anabolic pathways divergent (Fig. 2). Some pathways are cyclic:

one starting component of the pathway is regenerated in a series of reactions that converts another

starting component into a product.

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FIGURE 1 Energy relationships between catabolic and anabolic pathways.

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FIGURE 2 Three types of nonlinear metabolic pathways

Most of the reactions in living cells fall into one of five general categories: (1) oxidation-reductions; (2)

reactions that make or break carbon-carbon bonds; (3) internal rearrangements, isomerizations, and

eliminations; (4) group transfers; and (5) free radical reactions. Reactions within each general category

usually proceed by a limited set of mechanisms and often employ characteristic cofactors.

1. Oxidation-reduction reactions Carbon atoms encountered in biochemistry can exist in five oxidation states, depending on the elements with which carbon shares electrons (Fig. 3). In many biological oxidations, a compound loses two electrons and two hydrogen ions (that is, two hydrogen atoms); these reactions are commonly called dehydrogenations and the enzymes that catalyze them are called dehydrogenases (Fig. 4). In some, but not all, biological oxidations, a carbon atom becomes covalently bonded to an oxygen atom. The enzymes that catalyze these oxidations are generally called oxidases or, if the oxygen atom is derived directly from molecular oxygen (O₂), oxygenases.

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FIGURE 3 The oxidation states of carbon in biomolecules

FIGURE 4 An oxidation-reduction reaction.

2. Reactions that make or break carbon-carbon bonds Heterolytic cleavage of a C-C bond yields carbanion and a carbocation (Fig. 5). Conversely, the formation of a C-C bond involves the combination of a nucleophilic carbanion and an electrophilic carbocation.

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FIGURE 5 Two mechanisms for cleavage of a C-C or C-H bond.

3. Internal rearrangements, isomerizations, and eliminations

Another common type of cellular reaction is an intramolecular rearrangement, in which redistribution of

electrons results in isomerization, transposition of double bonds, or cis-trans rearrangements of double bonds.

4. Group transfer reactions

The transfer of acyl, glycosyl, and phosphoryl groups from one nucleophile to another is common in living cells. Acyl group transfer generally involves the addition of a nucleophile to the carbonyl

carbon of an acyl group to form a tetrahedral intermediate.

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5. Free radical reactions Once thought to be rare, the homolytic cleavage of covalent bonds to generate free radicals has now been found in a range of biochemical processes.

As we undertake our study of intermediary metabolism, a final word. Keep in mind that the myriad reactions described in these pages take place in, and play crucial roles in, living organisms. As you encounter each reaction and each pathway ask, What does this chemical transformation do for the organism? How does this pathway interconnect with the other pathways operating simultaneously in

the same cell to produce the energy and products required for cell maintenance and growth? How do

the multilayered regulatory mechanisms cooperate to balance metabolic and energy inputs and outputs,

achieving the dynamic steady state of life? Studied with this perspective, metabolism provides fascinating and revealing insights into life, with countless applications in medicine, agriculture, environmental toxicology and biotechnology.

GLYCOLYSIS, GLUCONEOGENESIS, AND THE PENTOSE PHOSPHATE PATHWAY

Glycolysis

In glycolysis(from the Greek glykys, meaning “sweet,” and lysis, meaning “splitting”), a molecule of glucose is degraded in a series of enzyme-catalyzed reactions to yield two molecules of the three-carbon

compound pyruvate. During the sequential reactions of glycolysis, some of the free energy released from glucose is conserved in the form of ATP and NADH. Glycolysis was the first metabolic pathway to

be elucidated and is probably the best understood.

Fates of Pyruvate

With the exception of some interesting variations in the bacterial realm, the pyruvate formed by glycolysis is further metabolized via one of three catabolic routes. In aerobic organisms or tissues, under

aerobic conditions, glycolysis is only the first stage in the complete degradation of glucose.

Pyruvate is

oxidized, with loss of its carboxyl group as CO₂, to yield the acetyl group of acetyl-coenzyme A; the acetyl group is then oxidized completely to CO₂ by the citric acid cycle. The second route for pyruvate is

its reduction to lactate via lactic acid fermentation. The third major route of pyruvate catabolism leads

to ethanol.

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Glycogenolysis: Glycogen and Starch Are Degraded by Phosphorolysis

Glycogen in animal tissues and in microorganisms (and starch in plants) can be mobilized for use within the same cell by a phosphorolytic reaction catalyzed by glycogen phosphorylase (starch phosphorylase in plants). These enzymes catalyze an attack by P_i on the (α 1-4) glycosidic linkage that joins the last two glucose residues at a nonreducing end, generating glucose 1-phosphate and a polymer one glucose unit shorter. Phosphorolysis preserves some of the energy of the glycosidic bond in the phosphate ester glucose 1-phosphate. Glycogen phosphorylase (or starch phosphorylase) acts repetitively until it approaches an (α 1-6) branch point, where its action stops. A debranching enzyme removes the branches.

Glucose 1-phosphate produced by glycogen phosphorylase is converted to glucose 6-phosphate by phosphoglucomutase, which catalyzes the reversible reaction.

The glucose 6-phosphate thus formed can enter glycolysis or another pathway such as the pentose phosphate pathway.

Gluconeogenesis

The central role of glucose in metabolism arose early in evolution, and this sugar remains the nearly universal fuel and building block in modern organisms, from microbes to humans. In mammals, some tissues depend almost completely on glucose for their metabolic energy. For

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the human brain and nervous system, as well as the erythrocytes, testes, renal medulla, and embryonic tissues, glucose from the blood is the sole or major fuel source. The brain alone requires about 120 g of glucose each day—more than half of all the glucose stored as glycogen in muscle and liver. However, the supply of glucose from these stores is not always sufficient; between meals and during longer fasts, or after vigorous exercise, glycogen is depleted. For these times, organisms need a method for synthesizing glucose from noncarbohydrate precursors. This is accomplished by a pathway called gluconeogenesis (“formation of new sugar”), which converts pyruvate and related three- and four-carbon compounds to glucose.

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Pentose Phosphate Pathway of Glucose Oxidation

Of particular importance in some tissues is the oxidation of glucose 6-phosphate to pentose phosphates by the pentose phosphate pathway (also called the phosphogluconate pathway or the hexose monophosphate pathway).

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THE CITRIC ACID CYCLE

We consider first the conversion of pyruvate to acetyl groups, then the entry of those groups into the citric acid cycle, called the tricarboxylic acid (TCA) cycle or the Krebs cycle (after its

discoverer, Hans Krebs).

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Pyruvate Is Oxidized to Acetyl-CoA and CO₂

The overall reaction catalyzed by the pyruvate dehydrogenase (PDH) complex is an oxidative decarboxylation, an irreversible oxidation process in which the carboxyl group is removed from pyruvate as a molecule of CO₂.

The PDH complex contains three enzymes—pyruvate dehydrogenase (E1), dihydrolipoyl transacetylase (E2), and dihydrolipoyl dehydrogenase (E3)—each present in multiple copies. The number of copies of each enzyme and therefore the size of the complex varies among species.

Reactions of the Citric Acid Cycle

We are now ready to trace the process by which acetyl-CoA undergoes oxidation. This chemical transformation is carried out by the citric acid cycle, the first cyclic pathway we have encountered.

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Stoichiometry of Coenzyme Reduction and ATP Formation in the Aerobic Oxidation of Glucose via Glycolysis, the Pyruvate Dehydrogenase Complex Reaction, the Citric Acid Cycle, and Oxidative Phosphorylation

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FATTY ACID CATABOLISM

The oxidation of long-chain fatty acids to acetyl-CoA is a central energy-yielding pathway in many organisms and tissues. In mammalian heart and liver, for example, it provides as much as 80% of the

energetic needs under all physiological circumstances. The electrons removed from fatty acids during

oxidation pass through the respiratory chain, driving ATP synthesis; the acetyl-CoA produced from the

fatty acids may be completely oxidized to CO₂ in the citric acid cycle, resulting in further energy conservation. In some species and in some tissues, the acetyl-CoA has alternative fates.

In liver, acetyl-CoA may be converted to ketone bodies—water-soluble fuels exported to the brain and

other tissues when glucose is not available. In higher plants, acetyl-CoA serves primarily as a biosynthetic precursor, only secondarily as fuel. Although the biological role of fatty acid oxidation differs from organism to organism, the mechanism is essentially the same. The repetitive four-step process, called oxidation, by which fatty acids are converted into acetyl-CoA is the main topic of this

module.

Digestion, Mobilization, and Transport of Fats

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Hormones Trigger Mobilization of Stored Triacylglycerols

When low levels of glucose in the blood trigger the release of glucagon,

1 the hormone binds its receptor in the adipocyte membrane and thus

2 stimulates adenylyl cyclase, via a G protein, to produce cAMP. This activates PKA, which phosphorylates

3 the hormone-sensitive lipase and

4 perilipin molecules on the surface of the lipid droplet. Phosphorylation of perilipin permits hormonesensitive lipase access to the surface of the lipid droplet, where

5 it hydrolyzes triacylglycerols to free fatty acids.

6 Fatty acids leave the adipocyte, bind serum albumin in the blood, and are carried in the blood; they

are released from the albumin and

7 enter a myocyte via a specific fatty acid transporter.

8 In the myocyte, fatty acids are oxidized to CO₂, and the energy of oxidation is conserved in ATP, which

fuels muscle contraction and other energy requiring metabolism in the myocyte.

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About 95% of the biologically available energy of triacylglycerols resides in their three long-chain fatty

acids; only 5% is contributed by the glycerol moiety. The glycerol released by lipase action is phosphorylated by glycerol kinase, and the resulting glycerol 3-phosphate is oxidized to dihydroxyacetone phosphate. The glycolytic enzyme triose phosphate isomerase converts this compound to glyceraldehyde 3-phosphate, which is oxidized via glycolysis.

Fatty Acids Are Activated and Transported into Mitochondria

The enzymes of fatty acid oxidation in animal cells are located in the mitochondrial matrix, as demonstrated in 1948 by Eugene P. Kennedy and Albert Lehninger.

Conversion of a fatty acid to a fatty acyl-CoA.

The conversion is catalyzed by fatty acyl-CoA synthetase and inorganic pyrophosphatase. Fatty acid

activation by formation of the fatty acyl-CoA derivative occurs in two steps. In step 1, the carboxylate

ion displaces the outer two phosphates of ATP to form a fatty acyl-adenylate, the mixed anhydride of a

carboxylic acid and a phosphoric acid. The other product is PP_i, an excellent leaving group that is

immediately hydrolyzed to two Pi, pulling the reaction in the forward direction.

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In step 2, the thiol group of coenzyme A carries out nucleophilic attack on the enzyme-bound mixed anhydride, displacing AMP and forming the thioester fatty acyl-CoA. The overall reaction is highly exergonic.

Fatty acids destined for mitochondrial oxidation are transiently attached to the hydroxyl group of carnitine to form fatty acyl-carnitine—the second reaction of the shuttle.

After fatty acyl-carnitine is formed at the outer membrane or in the intermembrane space, it moves into the matrix by facilitated diffusion through the transporter in the inner membrane. In the matrix, the

acyl group is transferred to mitochondrial coenzyme A, freeing carnitine to return to the intermembrane

space through the same transporter. Acyltransferase I is inhibited by malonyl-CoA, the first intermediate

in fatty acid synthesis. This inhibition prevents the simultaneous synthesis and degradation of fatty acids.

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Oxidation of Fatty Acids

β -oxidation—fatty acids undergo oxidative removal of successive two-carbon units in the form of acetyl-

CoA, starting from the carboxyl end of the fatty acyl chain. For example, the 16-carbon palmitic acid

(palmitate at pH 7) undergoes seven passes through the oxidative sequence, in each pass losing two

carbons as acetyl-CoA. At the end of seven cycles the last two carbons of palmitate (originally C-15 and

C-16) remain as acetyl-CoA. The overall result is the conversion of the 16-carbon chain of palmitate to

eight two-carbon acetyl groups of acetyl-CoA molecules.

The β -oxidation pathway.

(a) In each pass through this four-step sequence, one acetyl residue (shaded in pink) is removed in the

form of acetyl-CoA from the carboxyl end of the fatty acyl chain—in this example palmitate (C16), which

enters as palmitoyl-CoA. (b) Six more passes through the pathway yield seven more molecules of acetyl-

CoA, the seventh arising from the last two carbon atoms of the 16-carbon chain. Eight molecules of acetyl-CoA are formed in all.

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Oxidation of a monounsaturated fatty acid.

Oleic acid, as oleoyl-CoA (9), is the example used here. Oxidation requires an additional enzyme, enoyl-

CoA isomerase, to reposition the double bond, converting the cis isomer to a trans isomer, a normal intermediate in oxidation.

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Oxidation of a polyunsaturated fatty acid.

The example here is linoleic acid, as linoleoyl-CoA (9,12). Oxidation requires a second auxiliary enzyme

in addition to enoyl-CoA isomerase: NADPH-dependent 2,4-dienoyl-CoA reductase. The combined action

of these two enzymes converts a trans-2,cis-4-dienoyl-CoA intermediate to the trans-2-enoyl-CoA substrate necessary for oxidation.

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β -oxidation of odd-number fatty acids.

The sequence involves the carboxylation of propionyl-CoA to D-methylmalonyl-CoA and conversion of

the latter to succinyl-CoA. This conversion requires epimerization of D- to L-methylmalonyl-CoA, followed by a remarkable reaction in which substituents on adjacent carbon atoms exchange positions.

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Yield of ATP during Oxidation of One Molecule of Palmitoyl-CoA to CO₂ and H₂O

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Biosynthesis of Fatty Acids

After the discovery that fatty acid oxidation takes place by the oxidative removal of successive two-carbon (acetyl-CoA) units, biochemists thought the biosynthesis of fatty acids might proceed by a simple

reversal of the same enzymatic steps. However, as they were to find out, fatty acid biosynthesis and

breakdown occur by different pathways, are catalyzed by different sets of enzymes, and take place in

different parts of the cell. Moreover, biosynthesis requires the participation of a three-carbon intermediate, malonyl-CoA, that is not involved in fatty acid breakdown.

The formation of malonyl-CoA from acetyl-CoA is an irreversible process, catalyzed by acetyl-CoA

carboxylase.

Acetyl-CoA carboxylase has three functional regions: biotin carrier protein; biotin carboxylase, which activates CO₂ by attaching it to a nitrogen in the biotin ring in an ATP-dependent reaction; and transcarboxylase, which transfers activated CO₂ from biotin to acetyl-CoA, producing malonyl-CoA. The long, flexible biotin arm carries the activated CO₂ from the biotin carboxylase region to the transcarboxylase active site. The active enzyme in each step is shaded.

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Fatty Acid Synthesis

All the reactions in the synthetic process are catalyzed by a multienzyme complex, fatty acid synthase.

Each malonyl group and acetyl (or longer acyl) group is activated by a thioester that links it to fatty acid

synthase, a multienzyme complex.

1 Condensation of an activated acyl group (an acetyl group from acetyl-CoA is the first acyl group) and

two carbons derived from malonyl-CoA, with elimination of CO₂ from the malonyl group, extends the

acyl chain by two carbons. The mechanism of the first step of this reaction is given to illustrate the role

of decarboxylation in facilitating condensation. The β -keto product of this condensation is then reduced

in three more steps nearly identical to the reactions of oxidation, but in the reverse sequence:

2 the β -keto group is reduced to an alcohol,

3 elimination of H₂O creates a double bond, and

4 the double bond is reduced to form the corresponding saturated fatty acyl group.

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AMINO ACID OXIDATION

In animals, amino acids undergo oxidative degradation in three different metabolic circumstances:

1. During the normal synthesis and degradation of cellular proteins, some amino acids that are released

from protein breakdown and are not needed for new protein synthesis undergo oxidative degradation.

2. When a diet is rich in protein and the ingested amino acids exceed the body's needs for protein synthesis, the surplus is catabolized; amino acids cannot be stored.

3. During starvation or in uncontrolled diabetes mellitus, when carbohydrates are either unavailable or not properly utilized, cellular proteins are used as fuel.

Under all these metabolic conditions, amino acids lose their amino groups to form α -keto acids, the “carbon skeletons” of amino acids. The α -keto acids undergo oxidation to CO_2 and H_2O or, often more importantly, provide three- and four-carbon units that can be converted by gluconeogenesis into glucose, the fuel for brain, skeletal muscle, and other tissues.

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Metabolic Fates of Amino Groups

Amino acids derived from dietary protein are the source of most amino groups. Most amino acids are metabolized in the liver. Some of the ammonia generated in this process is recycled and used in a variety of biosynthetic pathways; the excess is either excreted directly or converted to urea or uric acid for excretion, depending on the organism. Excess ammonia generated in other (extrahepatic) tissues travels to the liver (in the form of amino groups, as described below) for conversion to the excretory form.

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Enzyme-catalyzed transaminations

The first step in the catabolism of most L-amino acids, once they have reached the liver, is removal of the α -amino groups, promoted by enzymes called aminotransferases or transaminases. In these transamination reactions, the L-amino group is transferred to the α -carbon atom of α -ketoglutarate, leaving behind the corresponding α -keto acid analog of the amino acid.

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Ammonia transport in the form of glutamine

Excess ammonia in tissues is added to glutamate to form glutamine, a process catalyzed by glutamine synthetase. After transport in the bloodstream, the glutamine enters the liver and NH_4^+ is liberated in mitochondria by the enzyme glutaminase.

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Urea cycle

The enzymes catalyzing these reactions (named in the text) are distributed between the mitochondrial matrix and the cytosol. One amino group enters the urea cycle as carbamoyl phosphate, formed in the matrix; the other enters as aspartate, formed in the matrix by transamination of oxaloacetate and glutamate, catalyzed by aspartate aminotransferase. The urea cycle consists of four steps.

- 1 Formation of citrulline from ornithine and carbamoyl phosphate (entry of the first amino group); the citrulline passes into the cytosol.
- 2 Formation of argininosuccinate through a citrullyl-AMP intermediate (entry of the second amino group).
- 3 Formation of arginine from argininosuccinate; this reaction releases

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fumarate, which enters the citric acid cycle. 4 Formation of urea; this reaction also regenerates, ornithine.

Summary of amino acid catabolism

Amino acids are grouped according to their major degradative end product. Some amino acids are listed

more than once because different parts of their carbon skeletons are degraded to different end products.

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Biosynthesis of the Nutritionally Nonessential Amino Acids

The enzymes glutamate dehydrogenase, glutamine synthetase, and aminotransferases occupy central

positions in amino acid biosynthesis. The combined effect of those three enzymes is to transform ammonium ion into the α -amino nitrogen of various amino acids.

Glutamate and Glutamine.

Reductive amination of α -ketoglutarate is catalyzed by glutamate dehydrogenase. Amination of glutamate to glutamine is catalyzed by glutamine synthetase. Alanine. Transamination of pyruvate forms alanine.

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Aspartate and Asparagine.

Transamination of oxaloacetate forms aspartate. The conversion of aspartate to asparagine is catalyzed

by asparagine synthetase.

Metabolism of Xenobiotics

Increasingly, humans are subjected to exposure to various foreign chemicals (xenobiotics)—drugs, food additives, pollutants, etc. The situation is well summarized in the following quotation from Rachel Carson: “As crude a weapon as the cave man’s club, the chemical barrage has been hurled against the fabric of life.” Understanding how xenobiotics are handled at the cellular level is important in learning how to cope with the chemical onslaught.

A xenobiotic (Gk xenos “stranger”) is a compound that is foreign to the body. The principal classes of xenobiotics of medical relevance are drugs, chemical carcinogens, and various compounds that have found their way into our environment by one route or another, such as polychlorinated biphenyls (PCBs) and certain insecticides. More than 200,000 manufactured environmental chemicals exist. Most of these

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Paper 4 of 4 — EMT 221

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eliminations; (4) group transfers; and (5) free radical reactions. Reactions within each general category

usually proceed by a limited set of mechanisms and often employ characteristic cofactors.

1. Oxidation-reduction reactions Carbon atoms encountered in biochemistry can exist in five oxidation states, depending on the elements with which carbon shares electrons (Fig. 3). In many biological oxidations, a compound loses two electrons and two hydrogen ions (that is, two hydrogen atoms); these reactions are commonly called dehydrogenations and the enzymes that catalyze them are called dehydrogenases (Fig. 4). In some, but not all, biological oxidations, a carbon atom becomes covalently bonded to an oxygen atom. The enzymes that catalyze these oxidations are generally called oxidases or, if the oxygen atom is derived directly from molecular oxygen (O₂), oxygenases.

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FIGURE 3 The oxidation states of carbon in biomolecules

FIGURE 4 An oxidation-reduction reaction.

2. Reactions that make or break carbon-carbon bonds Heterolytic cleavage of a C-C bond yields carbanion and a carbocation (Fig. 5). Conversely, the formation of a C-C bond involves the combination of a nucleophilic carbanion and an electrophilic carbocation.

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FIGURE 5 Two mechanisms for cleavage of a C-C or C-H bond.

3. Internal rearrangements, isomerizations, and eliminations

Another common type of cellular reaction is an intramolecular rearrangement, in which redistribution of

electrons results in isomerization, transposition of double bonds, or cis-trans rearrangements of double bonds.

4. Group transfer reactions

The transfer of acyl, glycosyl, and phosphoryl groups from one nucleophile to another is common in living cells. Acyl group transfer generally involves the addition of a nucleophile to the carbonyl

carbon of an acyl group to form a tetrahedral intermediate.

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5. Free radical reactions Once thought to be rare, the homolytic cleavage of covalent bonds to generate free radicals has now been found in a range of biochemical processes.

As we undertake our study of intermediary metabolism, a final word. Keep in mind that the myriad reactions described in these pages take place in, and play crucial roles in, living organisms. As you encounter each reaction and each pathway ask, What does this chemical transformation do for the organism? How does this pathway interconnect with the other pathways operating simultaneously in

the same cell to produce the energy and products required for cell maintenance and growth? How do

the multilayered regulatory mechanisms cooperate to balance metabolic and energy inputs and outputs,

achieving the dynamic steady state of life? Studied with this perspective, metabolism provides fascinating and revealing insights into life, with countless applications in medicine, agriculture, environmental toxicology and biotechnology.

GLYCOLYSIS, GLUCONEOGENESIS, AND THE PENTOSE PHOSPHATE PATHWAY

Glycolysis

In glycolysis(from the Greek glykys, meaning “sweet,” and lysis, meaning “splitting”), a molecule of glucose is degraded in a series of enzyme-catalyzed reactions to yield two molecules of the three-carbon

compound pyruvate. During the sequential reactions of glycolysis, some of the free energy released from glucose is conserved in the form of ATP and NADH. Glycolysis was the first metabolic pathway to

be elucidated and is probably the best understood.

Fates of Pyruvate

With the exception of some interesting variations in the bacterial realm, the pyruvate formed by glycolysis is further metabolized via one of three catabolic routes. In aerobic organisms or tissues, under

aerobic conditions, glycolysis is only the first stage in the complete degradation of glucose.

Pyruvate is

oxidized, with loss of its carboxyl group as CO₂, to yield the acetyl group of acetyl-coenzyme A; the acetyl group is then oxidized completely to CO₂ by the citric acid cycle. The second route for pyruvate is

its reduction to lactate via lactic acid fermentation. The third major route of pyruvate catabolism leads

to ethanol.

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Glycogenolysis: Glycogen and Starch Are Degraded by Phosphorolysis

Glycogen in animal tissues and in microorganisms (and starch in plants) can be mobilized for use within the same cell by a phosphorolytic reaction catalyzed by glycogen phosphorylase (starch phosphorylase in plants). These enzymes catalyze an attack by P_i on the (α 1-4) glycosidic linkage that joins the last two glucose residues at a nonreducing end, generating glucose 1-phosphate and a polymer one glucose unit shorter. Phosphorolysis preserves some of the energy of the glycosidic bond in the phosphate ester glucose 1-phosphate. Glycogen phosphorylase (or starch phosphorylase) acts repetitively until it approaches an (α 1-6) branch point, where its action stops. A debranching enzyme removes the branches.

Glucose 1-phosphate produced by glycogen phosphorylase is converted to glucose 6-phosphate by phosphoglucomutase, which catalyzes the reversible reaction.

The glucose 6-phosphate thus formed can enter glycolysis or another pathway such as the pentose phosphate pathway.

Gluconeogenesis

The central role of glucose in metabolism arose early in evolution, and this sugar remains the nearly universal fuel and building block in modern organisms, from microbes to humans. In mammals, some tissues depend almost completely on glucose for their metabolic energy. For

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the human brain and nervous system, as well as the erythrocytes, testes, renal medulla, and embryonic tissues, glucose from the blood is the sole or major fuel source. The brain alone requires about 120 g of glucose each day—more than half of all the glucose stored as glycogen in muscle and liver. However, the supply of glucose from these stores is not always sufficient; between meals and during longer fasts, or after vigorous exercise, glycogen is depleted. For these times, organisms need a method for synthesizing glucose from noncarbohydrate precursors. This is accomplished by a pathway called gluconeogenesis (“formation of new sugar”), which converts pyruvate and related three- and four-carbon compounds to glucose.

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Pentose Phosphate Pathway of Glucose Oxidation

Of particular importance in some tissues is the oxidation of glucose 6-phosphate to pentose phosphates by the pentose phosphate pathway (also called the phosphogluconate pathway or the hexose monophosphate pathway).

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THE CITRIC ACID CYCLE

We consider first the conversion of pyruvate to acetyl groups, then the entry of those groups into the citric acid cycle, called the tricarboxylic acid (TCA) cycle or the Krebs cycle (after its

discoverer, Hans Krebs).

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Pyruvate Is Oxidized to Acetyl-CoA and CO₂

The overall reaction catalyzed by the pyruvate dehydrogenase (PDH) complex is an oxidative decarboxylation, an irreversible oxidation process in which the carboxyl group is removed from pyruvate as a molecule of CO₂.

The PDH complex contains three enzymes—pyruvate dehydrogenase (E1), dihydrolipoyl transacetylase (E2), and dihydrolipoyl dehydrogenase (E3)—each present in multiple copies. The number of copies of each enzyme and therefore the size of the complex varies among species.

Reactions of the Citric Acid Cycle

We are now ready to trace the process by which acetyl-CoA undergoes oxidation. This chemical transformation is carried out by the citric acid cycle, the first cyclic pathway we have encountered.

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Stoichiometry of Coenzyme Reduction and ATP Formation in the Aerobic Oxidation of Glucose via Glycolysis, the Pyruvate Dehydrogenase Complex Reaction, the Citric Acid Cycle, and Oxidative Phosphorylation

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FATTY ACID CATABOLISM

The oxidation of long-chain fatty acids to acetyl-CoA is a central energy-yielding pathway in many organisms and tissues. In mammalian heart and liver, for example, it provides as much as 80% of the

energetic needs under all physiological circumstances. The electrons removed from fatty acids during

oxidation pass through the respiratory chain, driving ATP synthesis; the acetyl-CoA produced from the

fatty acids may be completely oxidized to CO₂ in the citric acid cycle, resulting in further energy conservation. In some species and in some tissues, the acetyl-CoA has alternative fates.

In liver, acetyl-CoA may be converted to ketone bodies—water-soluble fuels exported to the brain and

other tissues when glucose is not available. In higher plants, acetyl-CoA serves primarily as a biosynthetic precursor, only secondarily as fuel. Although the biological role of fatty acid oxidation differs from organism to organism, the mechanism is essentially the same. The repetitive four-step process, called oxidation, by which fatty acids are converted into acetyl-CoA is the main topic of this

module.

Digestion, Mobilization, and Transport of Fats

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Hormones Trigger Mobilization of Stored Triacylglycerols

When low levels of glucose in the blood trigger the release of glucagon,

1 the hormone binds its receptor in the adipocyte membrane and thus

2 stimulates adenylyl cyclase, via a G protein, to produce cAMP. This activates PKA, which phosphorylates

3 the hormone-sensitive lipase and

4 perilipin molecules on the surface of the lipid droplet. Phosphorylation of perilipin permits hormonesensitive lipase access to the surface of the lipid droplet, where

5 it hydrolyzes triacylglycerols to free fatty acids.

6 Fatty acids leave the adipocyte, bind serum albumin in the blood, and are carried in the blood; they

are released from the albumin and

7 enter a myocyte via a specific fatty acid transporter.

8 In the myocyte, fatty acids are oxidized to CO₂, and the energy of oxidation is conserved in ATP, which

fuels muscle contraction and other energy requiring metabolism in the myocyte.

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About 95% of the biologically available energy of triacylglycerols resides in their three long-chain fatty

acids; only 5% is contributed by the glycerol moiety. The glycerol released by lipase action is phosphorylated by glycerol kinase, and the resulting glycerol 3-phosphate is oxidized to dihydroxyacetone phosphate. The glycolytic enzyme triose phosphate isomerase converts this compound to glyceraldehyde 3-phosphate, which is oxidized via glycolysis.

Fatty Acids Are Activated and Transported into Mitochondria

The enzymes of fatty acid oxidation in animal cells are located in the mitochondrial matrix, as demonstrated in 1948 by Eugene P. Kennedy and Albert Lehninger.

Conversion of a fatty acid to a fatty acyl-CoA.

The conversion is catalyzed by fatty acyl-CoA synthetase and inorganic pyrophosphatase. Fatty acid

activation by formation of the fatty acyl-CoA derivative occurs in two steps. In step 1, the carboxylate

ion displaces the outer two phosphates of ATP to form a fatty acyl-adenylate, the mixed anhydride of a

carboxylic acid and a phosphoric acid. The other product is PP_i, an excellent leaving group that is

immediately hydrolyzed to two Pi, pulling the reaction in the forward direction.

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In step 2, the thiol group of coenzyme A carries out nucleophilic attack on the enzyme-bound mixed anhydride, displacing AMP and forming the thioester fatty acyl-CoA. The overall reaction is highly exergonic.

Fatty acids destined for mitochondrial oxidation are transiently attached to the hydroxyl group of carnitine to form fatty acyl-carnitine—the second reaction of the shuttle.

After fatty acyl-carnitine is formed at the outer membrane or in the intermembrane space, it moves into the matrix by facilitated diffusion through the transporter in the inner membrane. In the matrix, the

acyl group is transferred to mitochondrial coenzyme A, freeing carnitine to return to the intermembrane

space through the same transporter. Acyltransferase I is inhibited by malonyl-CoA, the first intermediate

in fatty acid synthesis. This inhibition prevents the simultaneous synthesis and degradation of fatty acids.

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Oxidation of Fatty Acids

β -oxidation—fatty acids undergo oxidative removal of successive two-carbon units in the form of acetyl-

CoA, starting from the carboxyl end of the fatty acyl chain. For example, the 16-carbon palmitic acid

(palmitate at pH 7) undergoes seven passes through the oxidative sequence, in each pass losing two

carbons as acetyl-CoA. At the end of seven cycles the last two carbons of palmitate (originally C-15 and

C-16) remain as acetyl-CoA. The overall result is the conversion of the 16-carbon chain of palmitate to

eight two-carbon acetyl groups of acetyl-CoA molecules.

The β -oxidation pathway.

(a) In each pass through this four-step sequence, one acetyl residue (shaded in pink) is removed in the

form of acetyl-CoA from the carboxyl end of the fatty acyl chain—in this example palmitate (C16), which

enters as palmitoyl-CoA. (b) Six more passes through the pathway yield seven more molecules of acetyl-

CoA, the seventh arising from the last two carbon atoms of the 16-carbon chain. Eight molecules of acetyl-CoA are formed in all.

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Oxidation of a monounsaturated fatty acid.

Oleic acid, as oleoyl-CoA (9), is the example used here. Oxidation requires an additional enzyme, enoyl-

CoA isomerase, to reposition the double bond, converting the cis isomer to a trans isomer, a normal intermediate in oxidation.

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Oxidation of a polyunsaturated fatty acid.

The example here is linoleic acid, as linoleoyl-CoA (9,12). Oxidation requires a second auxiliary enzyme

in addition to enoyl-CoA isomerase: NADPH-dependent 2,4-dienoyl-CoA reductase. The combined action

of these two enzymes converts a trans-2,cis-4-dienoyl-CoA intermediate to the trans-2-enoyl-CoA substrate necessary for oxidation.

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β -oxidation of odd-number fatty acids.

The sequence involves the carboxylation of propionyl-CoA to D-methylmalonyl-CoA and conversion of

the latter to succinyl-CoA. This conversion requires epimerization of D- to L-methylmalonyl-CoA, followed by a remarkable reaction in which substituents on adjacent carbon atoms exchange positions.

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Yield of ATP during Oxidation of One Molecule of Palmitoyl-CoA to CO₂ and H₂O

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Biosynthesis of Fatty Acids

After the discovery that fatty acid oxidation takes place by the oxidative removal of successive two-carbon (acetyl-CoA) units, biochemists thought the biosynthesis of fatty acids might proceed by a simple

reversal of the same enzymatic steps. However, as they were to find out, fatty acid biosynthesis and

breakdown occur by different pathways, are catalyzed by different sets of enzymes, and take place in

different parts of the cell. Moreover, biosynthesis requires the participation of a three-carbon intermediate, malonyl-CoA, that is not involved in fatty acid breakdown.

The formation of malonyl-CoA from acetyl-CoA is an irreversible process, catalyzed by acetyl-CoA

carboxylase.

Acetyl-CoA carboxylase has three functional regions: biotin carrier protein; biotin carboxylase, which activates CO₂ by attaching it to a nitrogen in the biotin ring in an ATP-dependent reaction; and transcarboxylase, which transfers activated CO₂ from biotin to acetyl-CoA, producing malonyl-CoA. The long, flexible biotin arm carries the activated CO₂ from the biotin carboxylase region to the transcarboxylase active site. The active enzyme in each step is shaded.

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Fatty Acid Synthesis

All the reactions in the synthetic process are catalyzed by a multienzyme complex, fatty acid synthase.

Each malonyl group and acetyl (or longer acyl) group is activated by a thioester that links it to fatty acid

synthase, a multienzyme complex.

1 Condensation of an activated acyl group (an acetyl group from acetyl-CoA is the first acyl group) and

two carbons derived from malonyl-CoA, with elimination of CO₂ from the malonyl group, extends the

acyl chain by two carbons. The mechanism of the first step of this reaction is given to illustrate the role

of decarboxylation in facilitating condensation. The β -keto product of this condensation is then reduced

in three more steps nearly identical to the reactions of oxidation, but in the reverse sequence:

2 the β -keto group is reduced to an alcohol,

3 elimination of H₂O creates a double bond, and

4 the double bond is reduced to form the corresponding saturated fatty acyl group.

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AMINO ACID OXIDATION

In animals, amino acids undergo oxidative degradation in three different metabolic circumstances:

1. During the normal synthesis and degradation of cellular proteins, some amino acids that are released

from protein breakdown and are not needed for new protein synthesis undergo oxidative degradation.

2. When a diet is rich in protein and the ingested amino acids exceed the body's needs for protein synthesis, the surplus is catabolized; amino acids cannot be stored.

3. During starvation or in uncontrolled diabetes mellitus, when carbohydrates are either unavailable or not properly utilized, cellular proteins are used as fuel.

Under all these metabolic conditions, amino acids lose their amino groups to form α -keto acids, the “carbon skeletons” of amino acids. The α -keto acids undergo oxidation to CO_2 and H_2O or, often more importantly, provide three- and four-carbon units that can be converted by gluconeogenesis into glucose, the fuel for brain, skeletal muscle, and other tissues.

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Metabolic Fates of Amino Groups

Amino acids derived from dietary protein are the source of most amino groups. Most amino acids are metabolized in the liver. Some of the ammonia generated in this process is recycled and used in a variety of biosynthetic pathways; the excess is either excreted directly or converted to urea or uric acid for excretion, depending on the organism. Excess ammonia generated in other (extrahepatic) tissues travels to the liver (in the form of amino groups, as described below) for conversion to the excretory form.

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Enzyme-catalyzed transaminations

The first step in the catabolism of most L-amino acids, once they have reached the liver, is removal of the α -amino groups, promoted by enzymes called aminotransferases or transaminases. In these transamination reactions, the L-amino group is transferred to the α -carbon atom of α -ketoglutarate, leaving behind the corresponding α -keto acid analog of the amino acid.

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Ammonia transport in the form of glutamine

Excess ammonia in tissues is added to glutamate to form glutamine, a process catalyzed by glutamine synthetase. After transport in the bloodstream, the glutamine enters the liver and NH_4^+ is liberated in mitochondria by the enzyme glutaminase.

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Urea cycle

The enzymes catalyzing these reactions (named in the text) are distributed between the mitochondrial matrix and the cytosol. One amino group enters the urea cycle as carbamoyl phosphate, formed in the matrix; the other enters as aspartate, formed in the matrix by transamination of oxaloacetate and glutamate, catalyzed by aspartate aminotransferase. The urea cycle consists of four steps. 1 Formation of citrulline from ornithine and carbamoyl phosphate (entry of the first amino group); the citrulline passes into the cytosol. 2 Formation of argininosuccinate through a citrullyl-AMP intermediate (entry of the second amino group). 3 Formation of arginine from argininosuccinate; this reaction releases

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fumarate, which enters the citric acid cycle. 4 Formation of urea; this reaction also regenerates, ornithine.

Summary of amino acid catabolism

Amino acids are grouped according to their major degradative end product. Some amino acids are listed

more than once because different parts of their carbon skeletons are degraded to different end products.

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Biosynthesis of the Nutritionally Nonessential Amino Acids

The enzymes glutamate dehydrogenase, glutamine synthetase, and aminotransferases occupy central

positions in amino acid biosynthesis. The combined effect of those three enzymes is to transform ammonium ion into the α -amino nitrogen of various amino acids.

Glutamate and Glutamine.

Reductive amination of α -ketoglutarate is catalyzed by glutamate dehydrogenase. Amination of glutamate to glutamine is catalyzed by glutamine synthetase. Alanine. Transamination of pyruvate forms alanine.

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Aspartate and Asparagine.

Transamination of oxaloacetate forms aspartate. The conversion of aspartate to asparagine is catalyzed

by asparagine synthetase.

Metabolism of Xenobiotics

Increasingly, humans are subjected to exposure to various foreign chemicals (xenobiotics)—drugs, food additives, pollutants, etc. The situation is well summarized in the following quotation from Rachel Carson: “As crude a weapon as the cave man’s club, the chemical barrage has been hurled against the fabric of life.” Understanding how xenobiotics are handled at the cellular level is important in learning how to cope with the chemical onslaught.

A xenobiotic (Gk xenos “stranger”) is a compound that is foreign to the body. The principal classes of xenobiotics of medical relevance are drugs, chemical carcinogens, and various compounds that have found their way into our environment by one route or another, such as polychlorinated biphenyls (PCBs) and certain insecticides. More than 200,000 manufactured environmental chemicals exist. Most of these

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man and lion
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 of the digestive
 Comparison
 Man Lion
 Cow
 Omnivore Carnivore
 Mode of Herbivore
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small intestine, liver,

rectum, intestine,

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rectum,

anus.

More less

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Proteases, lipases,

Protease, amylases,

Enzyme

amylases, lipases,

Systems lipases, amylases,

disa  charidases,

nucleases.

disaccharidases.elastases,

pectinase.

phytase,

1

Microbial Bacteria such as Eseherichia coli,

Bacteroides,

symbiosis Fibrobacter, Clostridium,

Firmcutes,

Bacteroides,

Ruminococcus, Enterococcus. Fusobacterium,

Prevotella.

Protozoa Eubacterium,

such as Endodinium, Lactobacillus.,

Diplodinium.

Fungi

such

as Neocallimastix,

Piromyces.

Archaeosuch

as Methanobrevibacter.

Stomach acidic

Slightly Moderately

acidic acidic

Highly

acidity